

Paul Rongieras

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Research Interests

Self-Supervised Learning, Representation Learning, Neural Collapse.

Education

PhD in Machine Learning, Inria, Rennes 2026 – 2029

- "Distributional Perspective of Self-Supervised Learning". Supervised by Titouan Vayer (Inria) and Franck Picard (CNRS, ENS Lyon)

Diplôme de l'École Normale Supérieure Paris-Saclay 2022 – 2026

- 4-year research-oriented program with admission through national competitive examination
- **2025 – 2026: Master 2 MVA (Mathematics, Vision, Learning)**, ENS Paris-Saclay
Coursework: Deep Learning and Signal Processing, Computational Statistics, Machine Learning for Time Series, Statistical Learning with Extreme Values, Geometric Data Analysis, Representation Learning for Computer Vision, Advanced Deep Learning, Algorithms for Speech and NLP, Kernel Methods
- **2024 – 2025: Master 1 Mathematics and Statistics**, Université Paris Dauphine-PSL
Coursework: Optimization, Statistical Learning, Generalized Linear Models, Nonparametric Statistics, Monte-Carlo Methods, Control of Markov Chains, Poisson Processes, Time Series, Brownian Motion
- 2022 – 2024: Bachelor's third year and Master 1 in Quantitative Economics

Classe préparatoire ENS Paris-Saclay D2, Lycée Victor et Hélène Basch, Rennes 2020 – 2022

- Intensive 2-year program preparing for national competitive entrance examinations to French Grandes Écoles

Talks

Oral presentation, GdR IASIS workshop *Self-Supervised Visual Learning: Methods, Applications and Challenges*, Paris 15 October 2026

- *Understanding Dimensional Collapse in Self-Supervised Learning through Maximum Mean Discrepancy*

Research Experience

Machine Learning Research Intern, Inria, Rennes April 2026 – September 2026

- Studied dimensional collapse in self-supervised learning through a distributional perspective, under the supervision of Titouan Vayer (Inria) and Franck Picard (CNRS, ENS Lyon)

Master's Thesis in Applied Mathematics, Université Paris Dauphine-PSL [GitHub] 2024 – 2025

- Modeled career trajectories with semi-Markov processes, supervised by Madalina Olteanu (CEREMADE)

Other Experience

Data Science Intern, IGEDD (Ministère de la Transition Écologique) June 2025 – August 2025

Data Engineering Intern, INSEE Bretagne June 2024 – July 2024

Technical Skills

Programming: Python (PyTorch, NumPy, Pandas, scikit-learn, TensorFlow/Keras), R, SAS

Tools: Git, LaTeX, Jupyter, VS Code

Languages

French: Native **English**: Fluent (IELTS C1) **Spanish**: Intermediate (SIELE B1)